Name
Date
Course

Practice Test II (Version 2)

I) Write your name, the date, and the course title (i.e. Trigonometry).
II) Show all work.
III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) You will need a calculator for this test (be sure that it is in degree mode).
V) Each problem is worth four points for a total of 100 points.

Prove that the following equations are true by converting the left side to the right.

① $\frac{\csc(-B)}{\csc B} + 1 = 0$

② $\cos(A+360°) + \cos(A-360°) = 2\cos(-A)$

③ $\cot(90°-\theta)\cot\theta + \cot^2\theta = \csc^2\theta$

④ $\cos(6X) - \cos(14X)\cos(8X) = \sin(14X)\sin(8X)$

Simplify the following expressions. Write exact expressions. In other words, you do not have to use a calculator.

⑤ $\sin(3A)\cos(2A) - \sin(2A)\cos(3A)$

(6) $\tan A \sin A \cos A$

(7) $\frac{3}{\sec\theta} - \frac{2}{\sin\theta} + \frac{9}{\sec\theta}$

(8) $\dfrac{\frac{1}{\sin\beta}}{\cos^2\beta + \sin^2\beta}$

(9) Find the value of the expression below without a calculator.

$\csc 75^\circ$

Prove that each equation is true by converting the left side to the right.

(10) $\cot\theta - \tan\theta = \dfrac{\cos(2\theta)}{\sin\theta\cos\theta}$

(11) $\dfrac{\sin^2 w - 7\sin w + 10}{\sin^2 w - 25} = \dfrac{\sin w - 2}{\sin w + 5}$

(12) $\sin B(\csc B - \sin B) = \cos^2 B$

(13) Solve if $0 \le \alpha < 360°$.

$3 - \sin\left(\frac{\alpha}{2}\right) = 3\sin\left(\frac{\alpha}{2}\right) + 1$

(14) Solve if $0 \leq \theta \leq 360°$. Round to the nearest tenth.

$6\tan\theta - \cot\theta = 0$

(15) Solve if $0 < \beta < 360°$. Hint: Square both sides of the equation.

$\cot\beta + 1 = \csc\beta$

⑯ Solve if $0 < \alpha \leq 360°$. * There is no solution, but you must show why.

$$12\tan\alpha + 12\sec\alpha = 5\cos\alpha$$

⑰ Solve if $0 < \theta < 360°$.

$$\csc(2\theta + 90°) = -\sqrt{2}$$

⑱ Solve if $0 < W < 360°$.

$$\cos W + 10 = 4 - 5\cos W$$

Prove that the following equations are true by converting the left side to the right.

(19) $$\frac{6\cos A}{\sin(2A)} = 3\csc A$$

(20) $$\frac{\cos(2B) + 1 - \cos^2 B}{\sin^2 B} = \cot^2 B$$

(21) Solve if $0 \le A \le 360°$. Round to the nearest hundredth.

$$2 + 3\cos A = \cos(2A)$$

(22) If β is in standard position, and it terminates in QIV ($270° < \beta < 360°$), find the value of each expression below if $\cos\beta = 3/4$.

i) $\sin\left(\frac{\beta}{2}\right)$

ii) $\cot\left(\frac{\beta}{2}\right)$

㉓ If W is in standard position, and it terminates in QII ($90° < W < 180°$), find the value of each expression below if $\sin W = 12/13$.

i) $\cos\left(\frac{W}{2}\right)$

ii) $\sec\left(\frac{W}{2}\right)$

(24) If B is in standard position, and it terminates in QIII ($180° < B < 270°$) find the value of each expression below if $\sin B = -\frac{\sqrt{6}}{3}$.

i) $\cos(2B)$

ii) Find the value of $\sin(2B)$ using the value of $\cos(2B)$ that you found in part one of this problem.

Prove that the following equation is true by converting the left side to the right.

25. $$\frac{\cot^2\alpha - 1}{\cot\alpha - 1} = \frac{1}{\tan\alpha} + 1$$

Practice Test II (Version 2) Answers

① $\frac{\csc(-B)}{\csc B}+1=0$

$\frac{-\csc B}{\csc B}+1=$

$-1+1=$

$0=$

② $\cos(A+360°)+\cos(A-360°)=2\cos(-A)$

$\cos A+\cos A=$

$2\cos A=$

$2\cos(-A)=$

③ $\cot(90°-\theta)\cot\theta+\cot^2\theta=\csc^2\theta$

$\tan\theta\cot\theta+\cot^2\theta=$

$\tan\theta\left(\frac{1}{\tan\theta}\right)+\cot^2\theta=$

$1+\cot^2\theta=$

$\csc^2\theta=$

④ $\cos(6x)-\cos(14x)\cos(8x)=\sin(14x)\sin(8x)$

$\cos(14x-8x)-\cos(14x)\cos(8x)=$

$\cos(14x)\cos(8x)+\sin(14x)\sin(8x)-\cos(14x)\cos(8x)=$

$\sin(14x)\sin(8x)=$

⑤ $\boxed{\sin A}$

⑥ $\boxed{\sin^2 A}$

⑦ $\boxed{12\cos\theta-2\csc\theta}$

⑧ $\boxed{\csc\beta}$

⑨ $\boxed{\frac{2\sqrt{2}}{1+\sqrt{3}}}$

⑩ $\cot\theta-\tan\theta=\frac{\cos(2\theta)}{\sin\theta\cos\theta}$

$\frac{\cos\theta}{\sin\theta}-\frac{\sin\theta}{\cos\theta}=$

$\frac{\cos^2\theta}{\sin\theta\cos\theta}-\frac{\sin^2\theta}{\cos\theta\sin\theta}=$

$\frac{\cos^2\theta-\sin^2\theta}{\sin\theta\cos\theta}=$

$\frac{\cos(2\theta)}{\sin\theta\cos\theta}=$

(11) $\dfrac{\sin^2 W - 7\sin W + 10}{\sin^2 W - 25} = \dfrac{\sin W - 2}{\sin W + 5}$

$\dfrac{(\sin W - 2)(\sin W - 5)}{(\sin W - 5)(\sin W + 5)} =$

$\dfrac{\sin W - 2}{\sin W + 5} =$ ✓

(12) $\sin B(\csc B - \sin B) = \cos^2 B$

$\sin B \csc B - \sin^2 B =$

$\sin B\left(\dfrac{1}{\sin B}\right) - \sin^2 B =$

$1 - \sin^2 B =$

$\sin^2 B + \cos^2 B - \sin^2 B =$

$\cos^2 B =$ ✓

(13) $\boxed{\alpha = 60°, 300°}$

(14) $\boxed{\theta \approx 22.2°, 157.8°, 202.2°, 337.8°}$

(15) $\boxed{\beta = 90°,\ \beta \neq 270°}$

(16) $\alpha \neq 270°$ When we check this solution in the original equation, we see that $\tan 270°$ and $\sec 270°$ are undefined.

$\boxed{\text{No Solution}}$

(17) $\boxed{\theta = 67.5°, 112.5°, 247.5°, 292.5°}$

(18) $\boxed{W = 180°}$

(19) $\dfrac{6\cos A}{\sin(2A)} = 3\csc A$

$\dfrac{6\cos A}{2\sin A\cos A} =$

$\dfrac{6}{2\sin A} =$

$\dfrac{3}{\sin A} =$

$\boxed{3\csc A} =$ ✓

(20) $\dfrac{\cos(2B) + 1 - \cos^2 B}{\sin^2 B} = \cot^2 B$

$\dfrac{2\cos^2 B - 1 + 1 - \cos^2 B}{\sin^2 B} =$

$\dfrac{\cos^2 B}{\sin^2 B} =$

$\boxed{\cot^2 B} =$ ✓

(21) $\boxed{A \approx 133.33°, 226.67°}$

(22) i) $\boxed{\dfrac{\sqrt{2}}{4}}$

Reference my previous courses.

ii) $\dfrac{-\sqrt{14}}{\sqrt{2}} = -\sqrt{\dfrac{14}{2}} = \boxed{-\sqrt{7}}$

(23) i) $\boxed{\dfrac{2\sqrt{13}}{13}}$ ii) $\boxed{\dfrac{\sqrt{13}}{2}}$

(24) i) $\boxed{-\dfrac{1}{3}}$ ii) $\boxed{\dfrac{2\sqrt{2}}{3}}$

(25) $\dfrac{\cot^2\alpha - 1}{\cot\alpha - 1} = \dfrac{1}{\tan\alpha} + 1$

$\dfrac{(\cot\alpha - 1)(\cot\alpha + 1)}{\cot\alpha - 1} =$

$\cot\alpha + 1 =$

$\dfrac{1}{\tan\alpha} + 1 =$ ✓